



FOOD FOR THOUGHT

Ryan Fernando



FOR MORE
STORIES

WHY SAUNA TRUMPS COLD THERAPY

Recent research suggests that while cold therapy may provide temporary relief, sauna therapy offers **superior long-term benefits**, making it the ultimate recovery tool in sport.

In the realm of post-workout recovery techniques, cold therapy has long been hailed as a go-to method for reducing inflammation, accelerating muscle recovery, and enhancing overall performance. From ice baths to cold showers, athletes and fitness enthusiasts have sworn by its benefits. However, recent scientific research suggests that cold therapy might not be as effective as its counterpart — sauna therapy. In this article, we delve into the comparative efficacy of cold therapy and sauna therapy, backed by scientific evidence.

Understanding cold therapy

Cold therapy involves exposing the body to cold temperatures, typically through methods like ice baths, cold plunges, or cold showers. Advocates of cold therapy argue that it constricts blood vessels, reduces inflammation, numbs nerve endings, and promotes recovery after strenuous physical activity.

Scientific evidence

While cold therapy indeed provides temporary relief and may help in reducing acute inflammation, its long-term benefits are questionable. A study published in the *European Journal of Applied*

Physiology compared the effects of cold water immersion (CWI) with active recovery on performance and inflammation markers after exhaustive exercise. Surprisingly, the researchers found that CWI did not enhance recovery or reduce inflammation compared to active recovery alone.

Moreover, another study published in the *Journal of Strength and Conditioning Research* examined the effects of cold water immersion on muscle soreness and strength recovery after resistance training. The results revealed that although cold water immersion reduced perceived soreness, it did not accelerate strength recovery when compared to passive recovery methods.

Limitations of cold therapy

One of the main limitations of cold therapy is its potential to impair muscle growth and adaptation. Cold exposure may interfere with the body's natural inflammatory response, which is crucial for muscle repair and growth. A study published in the *American Journal of Physiology — Cell Physiology* found that cold exposure inhibited muscle satellite cell activation, a key process in muscle repair and hypertrophy.

Furthermore, cold therapy's vasoconstrictive effects may hinder

nutrient delivery to muscles, thereby compromising the recovery process. Additionally, prolonged exposure to cold temperatures can lead to tissue damage and increase the risk of cold-related injuries, especially in individuals with underlying health conditions.

The rise of sauna

In contrast to cold therapy, sauna therapy involves exposing the body to high temperatures in a dry or wet sauna environment. Saunas induce sweating, increase heart rate, and promote vasodilation, leading to improved blood flow and circulation. Advocates of sauna therapy argue that it offers a myriad of health benefits, including enhanced cardiovascular function, improved endurance, and better recovery.

Scientific evidence

Numerous studies support the efficacy of sauna therapy in promoting recovery and enhancing athletic performance. A study published in the *Journal of Science and Medicine in Sport* investigated the effects of sauna bathing on endurance performance and recovery in competitive male runners. The researchers found that sauna bathing post-exercise improved time to exhaustion and increased plasma